

Chapter 5 Principal Components Analysis

Nguyễn Thị Mộng Ngọc
University of Science, VNU - HCM
ngtmngoc@hcmus.edu.vn

Principal Components - Description

- Data: p Variables observed on n Experimental/Sampling units
- Goal: Describe the Variance-Covariance structure of the variables through uncorrelated linear combinations of the variables
- Data Reduction: Fewer linear combinations (k) that “explain” large portion of total variation than the number of variables (p)
- Interpretation: New variables (principal components) can simplify the summary of the original variables.
- Used as input to other procedures such as Factor Analysis and Linear Regression with Highly Correlated Predictor Variables

Plan

- 1 Population Principal Components
 - Population Principal Components
 - Principal Components for Standardized Variables
 - Special Covariance Matrices
- 2 Principal Components with Sample Data
 - Choosing the Number of Principal Components to Retain
 - Standardizing Sample Principal Components
 - Large-Sample Inferences

Population Principal Components - I

p Random Variables: $X_1, \dots, X_p$

$$\mathbf{X} = \begin{bmatrix} X_1 \\ \vdots \\ X_p \end{bmatrix} \quad V\{\mathbf{X}\} = \mathbf{\Sigma} = \begin{bmatrix} \sigma_{11} & \cdots & \sigma_{1p} \\ \vdots & \ddots & \vdots \\ \sigma_{1p} & \cdots & \sigma_{pp} \end{bmatrix}$$

Eigenvalues of $\mathbf{\Sigma} \equiv \lambda_1 \geq \dots \geq \lambda_p \geq 0$.

Consider the linear combinations of $X_1, \dots, X_p$:

$$\begin{aligned} Y_1 &= a_{11}X_1 + a_{12}X_2 + \dots + a_{1p}X_p = \mathbf{a}_1'\mathbf{X} \\ Y_2 &= a_{21}X_1 + a_{22}X_2 + \dots + a_{2p}X_p = \mathbf{a}_2'\mathbf{X} \\ &\vdots \\ Y_p &= a_{p1}X_1 + a_{p2}X_2 + \dots + a_{pp}X_p = \mathbf{a}_p'\mathbf{X} \end{aligned}$$

Then, we have

$$V\{Y_k\} = V\{\mathbf{a}_k'\mathbf{X}\} = \sum_{i=1}^p a_{ki}^2 \sigma_{ii} + 2 \sum_{i=1}^{p-1} \sum_{i'=i+1}^p a_{ki} a_{ki'} \sigma_{ii'} = \mathbf{a}_k' \mathbf{\Sigma} \mathbf{a}_k$$

$$\text{COV}\{Y_i, Y_k\} = \text{COV}\{\mathbf{a}_i'\mathbf{X}, \mathbf{a}_k'\mathbf{X}\} = \mathbf{a}_i' \mathbf{\Sigma} \mathbf{a}_k \quad \text{with } i, k = 1, 2, \dots, p.$$

Population Principal Components - II

The Principal Components are those **uncorrelated** linear combination $Y_1, Y_2, \dots, Y_p$ whose **variances** $V\{Y_i\}, i = 1, 2, \dots, p$ are as large as possible.

- **First Principal Component** is linear combination $\mathbf{a}_1' \mathbf{X}$ that maximizes $V\{\mathbf{a}_1' \mathbf{X}\}$ s.t.

$$\mathbf{a}_1' \mathbf{a}_1 = 1$$

- **Second Principal Component** is linear combination $\mathbf{a}_2' \mathbf{X}$ that maximizes $V\{\mathbf{a}_2' \mathbf{X}\}$ s.t.

$$\mathbf{a}_2' \mathbf{a}_2 = 1 \quad \text{and} \quad \text{COV}\{\mathbf{a}_1' \mathbf{X}, \mathbf{a}_2' \mathbf{X}\} = \mathbf{a}_1' \boldsymbol{\Sigma} \mathbf{a}_2 = 0$$

- **k^{th} Principal Component** is linear combination $\mathbf{a}_k' \mathbf{X}$ that maximizes $V\{\mathbf{a}_k' \mathbf{X}\}$ s.t.

$$\mathbf{a}_k' \mathbf{a}_k = 1 \quad \text{and} \quad \text{COV}\{\mathbf{a}_i' \mathbf{X}, \mathbf{a}_k' \mathbf{X}\} = \mathbf{a}_i' \boldsymbol{\Sigma} \mathbf{a}_k = 0, \forall i < k.$$

Population Principal Components - III

$$\text{Random Vector } \mathbf{X} = \begin{bmatrix} X_1 \\ \vdots \\ X_p \end{bmatrix} \quad \text{with} \quad V\{\mathbf{X}\} = \mathbf{\Sigma} = \begin{bmatrix} \sigma_{11} & \cdots & \sigma_{1p} \\ \vdots & \ddots & \vdots \\ \sigma_{1p} & \cdots & \sigma_{pp} \end{bmatrix}$$

Eigenvalue/Eigenvector pairs of $\mathbf{\Sigma}$:

$$(\lambda_1, \mathbf{e}_1), \dots, (\lambda_p, \mathbf{e}_p) \quad \lambda_1 \geq \dots \geq \lambda_p \geq 0 \quad \mathbf{e}_i' \mathbf{e}_i = 1 \quad \mathbf{e}_i' \mathbf{e}_k = 0 \quad i \neq k$$

i^{th} Principal Component: $Y_i = \mathbf{e}_i' \mathbf{X} = e_{i1}X_1 + \dots + e_{ip}X_p \quad i = 1, \dots, p$

$$V\{Y_i\} = \mathbf{e}_i' \mathbf{\Sigma} \mathbf{e}_i = \mathbf{e}_i' \left[\sum_{i'=1}^p \lambda_{i'} \mathbf{e}_{i'} \mathbf{e}_{i'}' \right] \mathbf{e}_i = \lambda_i$$

$$\text{COV}\{Y_i, Y_k\} = \mathbf{e}_i' \mathbf{\Sigma} \mathbf{e}_k = \mathbf{e}_i' \left[\sum_{i'=1}^p \lambda_{i'} \mathbf{e}_{i'} \mathbf{e}_{i'}' \right] \mathbf{e}_k = 0 \quad i \neq k$$

Population Principal Components - IV

- $\max_{\mathbf{a} \neq 0} \frac{\mathbf{a}'\Sigma\mathbf{a}}{\mathbf{a}'\mathbf{a}} = \lambda_1$ attained at $\mathbf{a} = \mathbf{e}_1$

$$\Rightarrow \max_{\mathbf{a} \neq 0} \frac{\mathbf{a}'\Sigma\mathbf{a}}{\mathbf{a}'\mathbf{a}} = \lambda_1 = \frac{\mathbf{e}_1'\Sigma\mathbf{e}_1}{\mathbf{e}_1'\mathbf{e}_1} = \mathbf{e}_1'\Sigma\mathbf{e}_1 = V\{Y_1\}$$

- $\max_{\mathbf{a} \neq 0 \perp \mathbf{e}_1, \dots, \mathbf{e}_k} \frac{\mathbf{a}'\Sigma\mathbf{a}}{\mathbf{a}'\mathbf{a}} = \lambda_{k+1}$ attained at $\mathbf{a} = \mathbf{e}_{k+1}$ and $\mathbf{e}_{k+1}'\mathbf{e}_i = 0$

$$\Rightarrow \max_{\mathbf{a} \neq 0 \perp \mathbf{e}_1, \dots, \mathbf{e}_k} \frac{\mathbf{a}'\Sigma\mathbf{a}}{\mathbf{a}'\mathbf{a}} = \lambda_{k+1} = \frac{\mathbf{e}_{k+1}'\Sigma\mathbf{e}_{k+1}}{\mathbf{e}_{k+1}'\mathbf{e}_{k+1}} = \mathbf{e}_{k+1}'\Sigma\mathbf{e}_{k+1} = V\{Y_{k+1}\}$$

If all eigenvalues are distinct, all eigenvectors are orthogonal.

If eigenvalues not all distinct, eigenvectors corresponding to common eigenvalues can be chosen to be orthogonal

$$\Rightarrow \mathbf{e}_i'\mathbf{e}_k = 0 \quad i \neq k \quad \text{COV}\{Y_i, Y_k\} = \mathbf{e}_i'\Sigma\mathbf{e}_k = \mathbf{e}_i'\lambda_k\mathbf{e}_k = \lambda_k\mathbf{e}_i'\mathbf{e}_k = 0$$

Population Principal Components - V

$$\text{Random Vector } \mathbf{X} = \begin{bmatrix} X_1 \\ \vdots \\ X_p \end{bmatrix} \text{ with } V\{\mathbf{X}\} = \mathbf{\Sigma} = \begin{bmatrix} \sigma_{11} & \cdots & \sigma_{1p} \\ \vdots & \ddots & \vdots \\ \sigma_{1p} & \cdots & \sigma_{pp} \end{bmatrix}$$

Eigenvalue/Eigenvector pairs of $\mathbf{\Sigma}$:

$$(\lambda_1, \mathbf{e}_1), \dots, (\lambda_p, \mathbf{e}_p) \quad \lambda_1 \geq \dots \geq \lambda_p \geq 0 \quad \mathbf{e}_i' \mathbf{e}_i = 1 \quad \mathbf{e}_i' \mathbf{e}_k = 0 \quad i \neq k$$

$$i^{\text{th}} \text{ Principal Component: } Y_i = \mathbf{e}_i' \mathbf{X} = e_{i1}X_1 + \dots + e_{ip}X_p \quad i = 1, \dots, p$$

$$\sum_{i=1}^p V\{X_i\} = \sigma_{11} + \dots + \sigma_{pp} = \sum_{i=1}^p \sigma_{ii} = \text{trace}\{\mathbf{\Sigma}\}$$

$$\mathbf{\Sigma} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}' \quad \mathbf{P} = [\mathbf{e}_1 \quad \cdots \quad \mathbf{e}_p] \quad \mathbf{\Lambda} = \text{diag}\{\lambda_i\} \quad \mathbf{P}\mathbf{P}' = \mathbf{P}'\mathbf{P} = \mathbf{I}$$

$$\text{trace}\{\mathbf{\Sigma}\} = \text{trace}\{\mathbf{P}\mathbf{\Lambda}\mathbf{P}'\} = \text{trace}\{\mathbf{P}'\mathbf{P}\mathbf{\Lambda}\} = \text{trace}\{\mathbf{\Lambda}\} = \sum_{i=1}^p \lambda_i = \sum_{i=1}^p V\{Y_i\} = \sum_{i=1}^p V\{X_i\}$$

$$\text{Proportion of Total Population Variance due to } k^{\text{th}} \text{ Principal Component: } \frac{\lambda_k}{\lambda_1 + \dots + \lambda_p}$$

$k = 1, \dots, p.$

Population Principal Components - VI

$$\text{Random Vector } \mathbf{X} = \begin{bmatrix} X_1 \\ \vdots \\ X_p \end{bmatrix} \quad \text{with } V\{\mathbf{X}\} = \mathbf{\Sigma} = \begin{bmatrix} \sigma_{11} & \cdots & \sigma_{1p} \\ \vdots & \ddots & \vdots \\ \sigma_{1p} & \cdots & \sigma_{pp} \end{bmatrix}$$

Eigenvalue/Eigenvector pairs of $\mathbf{\Sigma}$:

$$(\lambda_1, \mathbf{e}_1), \dots, (\lambda_p, \mathbf{e}_p) \quad \lambda_1 \geq \dots \geq \lambda_p \geq 0 \quad \mathbf{e}_i' \mathbf{e}_i = 1 \quad \mathbf{e}_i' \mathbf{e}_k = 0 \quad i \neq k$$

$$i^{\text{th}} \text{ Principal Component: } Y_i = \mathbf{e}_i' \mathbf{X} = e_{i1}X_1 + \dots + e_{ip}X_p \quad i = 1, \dots, p$$

$$X_k = \mathbf{a}_k' \mathbf{X} \quad \mathbf{a}_k' = \begin{bmatrix} 0 & 1 & 0 \\ 1 \times (k-1) & & 1 \times (p-k) \end{bmatrix} \quad k = 1, \dots, p$$

$$\text{COV}\{X_k, Y_i\} = \text{COV}\{\mathbf{a}_k' \mathbf{X}, \mathbf{e}_i' \mathbf{X}\} = \mathbf{a}_k' \mathbf{\Sigma} \mathbf{e}_i = \mathbf{a}_k' \lambda_i \mathbf{e}_i = \lambda_i \mathbf{a}_k' \mathbf{e}_i = \lambda_i e_{ik}$$

$$V\{X_k\} = \sigma_{kk} \quad V\{Y_i\} = \lambda_i$$

$$\Rightarrow \text{CORR}\{X_k, Y_i\} = \rho_{Y_i, X_k} = \frac{\lambda_i e_{ik}}{\sqrt{\sigma_{kk}} \sqrt{\lambda_i}} = \frac{e_{ik} \sqrt{\lambda_i}}{\sqrt{\sigma_{kk}}}$$

Population Principal Components - VII

$$\text{Random Vector } \mathbf{X} = \begin{bmatrix} X_1 \\ \vdots \\ X_p \end{bmatrix} \text{ with } \mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$$

Density of $\mathbf{X}$ constant on $\boldsymbol{\mu}$ centered ellipsoids:

$$(\mathbf{X} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{X} - \boldsymbol{\mu}) = c^2 \text{ with axes } \pm c \sqrt{\lambda_i} \mathbf{e}_i \quad i = 1, \dots, p$$

$$\text{Set } \boldsymbol{\mu} = 0 \quad \mathbf{W} = \mathbf{X} - \boldsymbol{\mu} \Rightarrow E\{\mathbf{W}\} = 0 \quad V\{\mathbf{W}\} = \boldsymbol{\Sigma}$$

$$c^2 = \mathbf{X}' \boldsymbol{\Sigma}^{-1} \mathbf{X} = \frac{1}{\lambda_1} (\mathbf{e}_1' \mathbf{X})^2 + \dots + \frac{1}{\lambda_p} (\mathbf{e}_p' \mathbf{X})^2$$

Set $y_i = \mathbf{e}_i' \mathbf{X} \Rightarrow c^2 = \frac{1}{\lambda_1} y_1^2 + \dots + \frac{1}{\lambda_p} y_p^2$
 $y_1, \dots, y_p$ lie in directions of axes of constant density ellipsoid.

Principal Components for Standardized Variables - I

The standardized variables: $Z_1 = \frac{X_1 - \mu_1}{\sqrt{\sigma_{11}}}$, ..., $Z_p = \frac{X_p - \mu_p}{\sqrt{\sigma_{pp}}}$

$$\mathbf{Z} = \left(\mathbf{V}^{1/2} \right)^{-1} (\mathbf{X} - \boldsymbol{\mu}) \quad \text{where} \quad \mathbf{V}^{1/2} = \begin{bmatrix} \sqrt{\sigma_{11}} & 0 & \cdots & 0 \\ 0 & \sqrt{\sigma_{22}} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sqrt{\sigma_{pp}} \end{bmatrix}$$

$$E \{ \mathbf{Z} \} = E \left\{ \left(\mathbf{V}^{1/2} \right)^{-1} (\mathbf{X} - \boldsymbol{\mu}) \right\} = \left(\mathbf{V}^{1/2} \right)^{-1} (\boldsymbol{\mu} - \boldsymbol{\mu}) = \mathbf{0}$$

$$V \{ \mathbf{Z} \} = \left(\mathbf{V}^{1/2} \right)^{-1} \boldsymbol{\Sigma} \left(\mathbf{V}^{1/2} \right)^{-1} = \begin{bmatrix} \frac{\sigma_{11}}{\sqrt{\sigma_{11}}\sqrt{\sigma_{11}}} & \frac{\sigma_{12}}{\sqrt{\sigma_{11}}\sqrt{\sigma_{22}}} & \cdots & \frac{\sigma_{1p}}{\sqrt{\sigma_{11}}\sqrt{\sigma_{pp}}} \\ \frac{\sigma_{12}}{\sqrt{\sigma_{11}}\sqrt{\sigma_{22}}} & \frac{\sigma_{22}}{\sqrt{\sigma_{22}}\sqrt{\sigma_{22}}} & \cdots & \frac{\sigma_{2p}}{\sqrt{\sigma_{22}}\sqrt{\sigma_{pp}}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\sigma_{1p}}{\sqrt{\sigma_{11}}\sqrt{\sigma_{pp}}} & \frac{\sigma_{2p}}{\sqrt{\sigma_{22}}\sqrt{\sigma_{pp}}} & \cdots & \frac{\sigma_{pp}}{\sqrt{\sigma_{pp}}\sqrt{\sigma_{pp}}} \end{bmatrix} = \boldsymbol{\rho}.$$

Principal Components for Standardized Variables - II

Let $(\lambda_i, \mathbf{e}_i)$ be the Eigenvalue/Eigenvector pairs of $\boldsymbol{\rho}$ with

$$\lambda_1 \geq \dots \geq \lambda_p \geq 0$$

The i^{th} Principal Component of the standardized variables $\mathbf{Z} = \begin{bmatrix} Z_1 \\ \vdots \\ Z_p \end{bmatrix}$

where $V\{\mathbf{Z}\} = \boldsymbol{\rho}$ is:

$$Y_i = \mathbf{e}_i' \mathbf{Z} = \mathbf{e}_i' \left(\mathbf{V}^{1/2} \right)^{-1} (\mathbf{X} - \boldsymbol{\mu}) \quad i = 1, \dots, p$$

Moreover $\sum_{i=1}^p V\{Y_i\} = \sum_{i=1}^p V\{Z_i\} = p$ and $\rho_{Y_i, Z_k} = e_{ik} \sqrt{\lambda_i}$

Proportion of standardized population variance due to k^{th} Principal Component:

$$\frac{\lambda_k}{p} \quad k = 1, \dots, p.$$

Special Covariance Matrices - I

$$1) \text{ Independence: } \Sigma = \begin{bmatrix} \sigma_{11} & 0 & \cdots & 0 \\ 0 & \sigma_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_{pp} \end{bmatrix}$$

$\mathbf{e}_i = [0, \dots, 0, 1, 0, \dots, 0]'$ with 1 in the i th position

$$\lambda_1 = \max(\sigma_{11}, \dots, \sigma_{pp}) \quad \text{and} \quad \lambda_p = \min(\sigma_{11}, \dots, \sigma_{pp})$$

We observe that $\Sigma \mathbf{e}_i = \sigma_{ii} \mathbf{e}_i \quad \rho = 1 \quad \rho \mathbf{e}_i = 1 \mathbf{e}_i = \mathbf{e}_i$

$$\Sigma : (\sigma_{ii}, \mathbf{e}_i)$$

Eigenvalue/Eigenvector pairs:

$$\rho : (1, \mathbf{e}_i)$$

Special Covariance Matrices - II

2) Compound Symmetry:

$$\Sigma = \begin{bmatrix} \sigma^2 & \rho\sigma^2 & \cdots & \rho\sigma^2 \\ \rho\sigma^2 & \sigma^2 & \cdots & \rho\sigma^2 \\ \vdots & \vdots & \ddots & \vdots \\ \rho\sigma^2 & \rho\sigma^2 & \cdots & \sigma^2 \end{bmatrix} \quad \rho = \begin{bmatrix} 1 & \rho & \cdots & \rho \\ \rho & 1 & \cdots & \rho \\ \vdots & \vdots & \ddots & \vdots \\ \rho & \rho & \cdots & 1 \end{bmatrix}$$

$$\rho > 0 \Rightarrow \lambda_1 = 1 + (p-1)\rho \quad \lambda_2 = \dots = \lambda_p = 1 - \rho$$

$$\begin{aligned} \mathbf{e}_1' &= \left[\frac{1}{\sqrt{p}} \quad \frac{1}{\sqrt{p}} \quad \cdots \quad \frac{1}{\sqrt{p}} \right] & \mathbf{e}_2' &= \left[\frac{1}{\sqrt{1(2)}} \quad -\frac{1}{\sqrt{1(2)}} \quad 0 \quad \cdots \quad 0 \right] \\ \mathbf{e}_i' &= \left[\frac{1}{\sqrt{(i-1)i}} \quad \cdots \quad \frac{1}{\sqrt{(i-1)i}} \quad -\frac{(i-1)}{\sqrt{(i-1)i}} \quad 0 \quad \cdots \quad 0 \right] \\ \mathbf{e}_p' &= \left[\frac{1}{\sqrt{(p-1)p}} \quad \frac{1}{\sqrt{(p-1)p}} \quad \cdots \quad -\frac{(p-1)}{\sqrt{(p-1)p}} \right] \end{aligned}$$

$$Y_1 = \mathbf{e}_1' \mathbf{Z} = \sum_{i=1}^p \frac{1}{\sqrt{p}} Z_i = \frac{1}{\sqrt{p}} \sum_{i=1}^p Z_i \text{ Equal Weights} \quad \frac{\lambda_1}{p} = \frac{1+(p-1)\rho}{p} = \rho + \frac{1-\rho}{p}$$

If $\mathbf{Z} \sim \mathcal{N}(0, \rho)$ then Ellipsoids of Constant Density are "Cigar-Shaped "

with first P.C. as projection of $\mathbf{Z}$ on $\mathbf{1}' = [1 \quad \cdots \quad 1]$

Plan

1 Population Principal Components

- Population Principal Components
- Principal Components for Standardized Variables
- Special Covariance Matrices

2 Principal Components with Sample Data

- Choosing the Number of Principal Components to Retain
- Standardizing Sample Principal Components
- Large-Sample Inferences

Principal Components with Sample Data

$\mathbf{x}_1, \dots, \mathbf{x}_n \equiv$ observed random sample from population with $E\{\mathbf{X}\} = \boldsymbol{\mu}$, $V\{\mathbf{X}\} = \boldsymbol{\Sigma}$

→ Sample Mean: $\bar{\mathbf{x}}$, Sample Variance: $\mathbf{S}$, $\mathbf{R}$

$\mathbf{a}_1' \mathbf{x}_j = a_{11}x_{j1} + \dots + a_{1p}x_{jp} \quad j = 1, \dots, n$

→ Sample Mean: $\mathbf{a}_1' \bar{\mathbf{x}}$, Sample Variance: $\mathbf{a}_1' \mathbf{S} \mathbf{a}_1$, $\widehat{\text{COV}}\{\mathbf{a}_1' \mathbf{x}_j, \mathbf{a}_2' \mathbf{x}_j\} = \mathbf{a}_1' \mathbf{S} \mathbf{a}_2$.

If $\mathbf{S}_{p \times p}$ has eigenvalue/eigenvector pair $(\hat{\lambda}_1, \hat{\mathbf{e}}_1), \dots, (\hat{\lambda}_p, \hat{\mathbf{e}}_p)$ with $\hat{\lambda}_1 \geq \dots \geq \hat{\lambda}_p \geq 0$
the i th sample principal component is

$$\hat{y}_i = \hat{\mathbf{e}}_i' \mathbf{x} = \hat{\mathbf{e}}_{i1}x_1 + \dots + \hat{\mathbf{e}}_{ip}x_p \quad i = 1, \dots, p$$

$$\text{and } \widehat{V}\{\hat{y}_i\} = \hat{\lambda}_i \quad i = 1, \dots, p \quad \widehat{\text{COV}}\{\hat{y}_i, \hat{y}_k\} = 0 \quad i \neq k$$

Total Sample Variance:

$$\sum_{i=1}^p s_{ii} = \sum_{i=1}^p \hat{\lambda}_i \quad \text{and} \quad r_{\hat{y}_i, \hat{y}_k} = \frac{\hat{\mathbf{e}}_{ik} \sqrt{\hat{\lambda}_i}}{\sqrt{s_{kk}}} \quad i, k = 1, \dots, p$$

Sample Principal Components can be computed from $\hat{\boldsymbol{\Sigma}}$, $\mathbf{S}$, and $\mathbf{R}$

If data are centered the average of each P.C. is 0:

$$\hat{y}_{ji} = \hat{\mathbf{e}}_i' (\mathbf{x}_j - \bar{\mathbf{x}}) \Rightarrow \bar{\hat{y}}_i = \frac{1}{n} \sum_{j=1}^n \hat{y}_{ji} = 0$$

Choosing the Number of Principal Components to Retain

- Goal: Data Reduction $\Rightarrow$ Keeping $k < p$ Components
- PCA Based on Correlation Matrix $\Rightarrow$ Keep Components with Eigenvalues exceeding 1 (the average of the Eigenvalues)
- Scree Plot – Plot the Eigenvalues versus their component number and visually determine where curve flattens out
- Percentage of Variation Explained – If a given percentage is to be explained (say 75%), keep components that cover the threshold

Standardizing Sample Principal Components - I

Data Vector for j^{th} observed unit: $\mathbf{x}_j = \begin{bmatrix} x_{j1} \\ \vdots \\ x_{jp} \end{bmatrix} \quad j = 1, \dots, n$

$$\rightarrow \bar{\mathbf{x}} = \frac{1}{n} \sum_{j=1}^n \mathbf{x}_j = \begin{bmatrix} \bar{x}_1 \\ \vdots \\ \bar{x}_p \end{bmatrix} \quad \mathbf{S} = \frac{1}{n-1} \sum_{j=1}^n (\mathbf{x}_j - \bar{\mathbf{x}})(\mathbf{x}_j - \bar{\mathbf{x}})'$$

$$\mathbf{D}^{1/2} = \begin{bmatrix} \sqrt{s_{11}} & 0 & \cdots & 0 \\ 0 & \sqrt{s_{22}} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sqrt{s_{pp}} \end{bmatrix} \quad \mathbf{R} = \mathbf{D}^{-1/2} \mathbf{S} \mathbf{D}^{-1/2}$$

Standardizing Sample Principal Components - II

The $n \times p$ data matrix of standardized observations $\mathbf{Z}$

$$\mathbf{z}_j = \mathbf{D}^{-1/2} (\mathbf{x}_j - \bar{\mathbf{x}}) = \begin{bmatrix} \frac{x_{j1} - \bar{x}_1}{\sqrt{s_{11}}} \\ \vdots \\ \frac{x_{jp} - \bar{x}_p}{\sqrt{s_{pp}}} \end{bmatrix} \quad \mathbf{Z} = \begin{bmatrix} z_{11} & \cdots & z_{1p} \\ \vdots & \ddots & \vdots \\ z_{n1} & \cdots & z_{np} \end{bmatrix} = \begin{bmatrix} \mathbf{z}_1' \\ \vdots \\ \mathbf{z}_n' \end{bmatrix}$$

$$\bar{\mathbf{z}} = \frac{1}{n} \mathbf{Z}' \mathbf{1}_n = \begin{bmatrix} \bar{z}_1 \\ \vdots \\ \bar{z}_p \end{bmatrix} = \mathbf{0}$$

$$\begin{aligned} \mathbf{S}_z &= \frac{1}{n-1} (\mathbf{Z} - \frac{1}{n} \mathbf{1} \mathbf{1}' \mathbf{Z})' (\mathbf{Z} - \frac{1}{n} \mathbf{1} \mathbf{1}' \mathbf{Z}) = \frac{1}{n-1} (\mathbf{Z} - \mathbf{1} \bar{\mathbf{z}}')' (\mathbf{Z} - \mathbf{1} \bar{\mathbf{z}}') \\ &= \frac{1}{n-1} \mathbf{Z}' \mathbf{Z} = \mathbf{R} \end{aligned}$$

Note:

$$\sum_{j=1}^n z_{ji} z_{jk} = \sum_{j=1}^n \left[\left(\frac{x_{ji} - \bar{x}_i}{\sqrt{s_{ii}}} \right) \left(\frac{x_{jk} - \bar{x}_k}{\sqrt{s_{kk}}} \right) \right] = \frac{\sum_{j=1}^n (x_{ji} - \bar{x}_i)(x_{jk} - \bar{x}_k)}{\sqrt{s_{ii}} \sqrt{s_{kk}}} = \frac{(n-1) s_{ik}}{\sqrt{s_{ii}} \sqrt{s_{kk}}} = (n-1) r_{ik}.$$

Standardizing Sample Principal Components - III

$\mathbf{Z}_1, \dots, \mathbf{Z}_n \equiv$ standardized observations with Covariance matrix $\mathbf{R}$
 $(\hat{\lambda}_i, \hat{\mathbf{e}}_i) \equiv$ Eigenvalue/Eigenvector Pairs of $\mathbf{R}$ with $\hat{\lambda}_1 \geq \dots \geq \hat{\lambda}_p \geq 0$
 i^{th} Sample Principal Component:

$$\hat{y}_i = \hat{\mathbf{e}}_i' \mathbf{Z} = \hat{e}_{i1}z_1 + \dots + \hat{e}_{ip}z_p \quad i = 1, \dots, p$$

Sample Variance: $\widehat{V}\{\hat{y}_i\} = \hat{\lambda}_i \quad i = 1, \dots, p$

Sample Covariances: $\widehat{\text{COV}}\{\hat{y}_i, \hat{y}_k\} = 0 \quad i \neq k$

Total (Standardized) Sample Variance: $\text{trace}(\mathbf{R}) = p = \hat{\lambda}_1 + \dots + \hat{\lambda}_p$

$$\widehat{\text{CORR}}\{\hat{y}_i, z_k\} = r_{\hat{y}_i, z_k} = \hat{e}_{ik} \sqrt{\hat{\lambda}_i} \quad i, k = 1, \dots, p$$

Proportion of Standardized Sample Variance due to i^{th} Principal Component: $\frac{\hat{\lambda}_i}{p} \quad i = 1, \dots, p$

Graphical Methods

- Scatterplots of Units by their first few principal components allows to see which units are similar and different with respect to their primary component scores . Also can be used to assess normality
- Q-Q Plots to assess normality
- Biplot – Plot representing where experimental units and variables lie with respect to 2 principal components

Large-Sample Inferences - I

Assumptions: $\mathbf{X}_1, \dots, \mathbf{X}_n \equiv$ Random sample from Normal Population

Eigenvalues of Σ are distinct and positive $\lambda_1 > \dots > \lambda_p > 0$

Exception when testing for specific Covariance structure (e.g. Compound Symmetry)

Large - sample Theory for $\hat{\boldsymbol{\lambda}}' = \begin{bmatrix} \hat{\lambda}_1 & \dots & \hat{\lambda}_p \end{bmatrix}$ and $\hat{\mathbf{e}}_1, \dots, \hat{\mathbf{e}}_p$ from $\mathbf{S}$:

- 1) $\boldsymbol{\Lambda} = \text{diag} \{ \lambda_i \}$ from $\boldsymbol{\Sigma} \Rightarrow \sqrt{n} (\hat{\boldsymbol{\lambda}} - \boldsymbol{\lambda}) \approx \mathcal{N}_p (0, 2\boldsymbol{\Lambda}^2)$
- 2) $\mathbf{E}_i = \lambda_i \sum_{\substack{k=1 \\ k \neq i}}^p \frac{\lambda_k}{(\lambda_k - \lambda_i)^2} \mathbf{e}_k \mathbf{e}_k' \Rightarrow \sqrt{n} (\hat{\mathbf{e}}_i - \mathbf{e}_i) \approx \mathcal{N}_p (0, \mathbf{E}_i)$
- 3) $\{ \hat{\lambda}_i \} \perp \{ \hat{\mathbf{e}}_i \}$

Large-Sample Inferences - II

$$\begin{aligned}
 1) \quad \Rightarrow \quad 1 - \alpha &\approx P \left(-z \left(\frac{\alpha}{2} \right) \sqrt{2\lambda_i^2} \leq \sqrt{n} (\hat{\lambda}_i - \lambda_i) \leq z \left(\frac{\alpha}{2} \right) \sqrt{2\lambda_i^2} \right) \\
 &= P \left(-z \left(\frac{\alpha}{2} \right) \sqrt{\frac{2}{n}} \leq \left(\frac{\hat{\lambda}_i}{\lambda_i} - 1 \right) \leq z \left(\frac{\alpha}{2} \right) \sqrt{\frac{2}{n}} \right) = \\
 &= P \left(1 - z \left(\frac{\alpha}{2} \right) \sqrt{\frac{2}{n}} \leq \frac{\hat{\lambda}_i}{\lambda_i} \leq 1 + z \left(\frac{\alpha}{2} \right) \sqrt{\frac{2}{n}} \right) \\
 &= P \left(\frac{\hat{\lambda}_i}{1 + z \left(\frac{\alpha}{2} \right) \sqrt{\frac{2}{n}}} \leq \lambda_i \leq \frac{\hat{\lambda}_i}{1 - z \left(\frac{\alpha}{2} \right) \sqrt{\frac{2}{n}}} \right)
 \end{aligned}$$

Bonferroni method can be used for simultaneous CIs for m components using $z \left(\frac{\alpha}{2m} \right)$

2) For $\mathbf{E}_i$, $\{\hat{\lambda}_i\}$ and $\{\hat{\mathbf{e}}_i\}$ are used for estimated standard errors

Lawley's Test for Equal Correlation Structure

$$H_0 : \rho = \rho_0 = \begin{bmatrix} 1 & \rho & \cdots & \rho \\ \rho & 1 & \cdots & \rho \\ \vdots & \vdots & \ddots & \vdots \\ \rho & \rho & \cdots & 1 \end{bmatrix} \quad \text{vs} \quad H_A : \rho \neq \rho_0$$

$$\text{Total number of correlations} \equiv \binom{p}{2} = \frac{p(p-1)}{2}$$

$$\bar{r}_k = \frac{1}{p-1} \sum_{\substack{i=1 \\ i \neq k}}^p r_{ik} \quad \bar{r} = \frac{2}{p(p-1)} \sum_{i=1}^{p-1} \sum_{k=i+1}^p r_{ik} \quad \hat{\gamma} = \frac{(p-1)^2 [1 - (1-\bar{r})^2]}{p - (p-2)(1-\bar{r})^2}$$

with $k = 1, \dots, p$

$$TS : T = \frac{n-1}{(1-\bar{r})^2} \left[\sum_{i=1}^{p-1} \sum_{k=i+1}^p (r_{ik} - \bar{r})^2 - \hat{\gamma} \sum_{k=1}^p (\bar{r}_k - \bar{r})^2 \right]$$

$$RR : T \geq \chi^2_{(p+1)(p-2)/2}$$